

A POLYHEDRAL APPROACH FOR STOCHASTIC CHOICE MODELS

Sheng Cheng

November 29, 2025

1 INTRODUCTION

Stochastic choice models, which describe the randomness in consumer decisions, are a core subject in economics and operations management. A cornerstone of this literature, the random utility model (RUM), was formally introduced by [Block and Marschak, 1959] and has since become a foundational paradigm. In RUM, the consumer population is a mixture of heterogeneous consumer types, each of whom has a different rank of choices and chooses the alternative with highest rank in the offered menu. Then the natural problem is to axiomatize random utility model, that is, to characterize the choice behavior that can be explained by the random utility model. This problem is first solved by [Falmagne, 1978]. In his paper, Falmagne defines a Block-Marschak (B-M) polynomial and proves that a choice function admits the random utility representation if and only if its Block-Marschak polynomial is nonnegative. However, his proof is purely algebraic and lack of intuitive explanation. Later, [Fiorini, 2004] simplified his proof, using a polytope approach. In short, Fiorini builds the connection between random utility model with a network polytope, representing the ranking of choices as the path in the network.

A polytope approach has gained traction in the research of stochastic choice. A growing body of literature employs this approach to analyze the properties of existing choice models and to develop new ones. For example, [Chambers et al., 2024] used this approach to study their correlated choice model; [Turansick, 2022] used network graph to study the identification of random utility model. Another stream of research is dedicated to the mathematical and geometric properties of the choice polytope itself. A representative work is [Doignon and Saito, 2023], which characterizes the facets of the random utility flow polytope and the adjacencies among its vertices and facets.

1.1 OUR WORK

In the first part of this note, we follow Fiorini's technique to give a detailed proof of the axiomatization theorem of random utility model. Since Fiorini omitted some necessary arguments (for example, the flow balance result 3.2), we provide more details to serve as a complement to his proof.

In particular, we define three central polytopes for our analysis: the rational choice polytope $\mathcal{P}_{RC}$, the choice flow polytope $\mathcal{P}_{CF}$ and the Block-Marschak polytope $\mathcal{P}_{BM}$. We first translate the axiomatization of the RUM into the characterization of $\mathcal{H}$ -representation of $\mathcal{P}_{RC}$. To this end, we introduce the choice flow polytope. A key step is that we establish an invertible linear transformation f such that $f(\mathcal{P}_{RC}) = \mathcal{P}_{CF}$. Then we apply known results from combinatorial optimization, which provide the $\mathcal{H}$ -representation of the flow polytope. This representation is precisely the set of Block-Marschak inequalities that define $\mathcal{P}_{BM}$. Thereby, we complete the axiomatization theorem of random utility using a polyhedral approach.

In the second part of this note, we apply the polyhedral approach to analyze other stochastic choice models. These include the branching-independent random utility model [Suleymanov, 2024], the multinomial logit model (MNL), and a new model—the partial set-dependent random utility model—adapted from [Feng et al., 2023]. Our analysis proceeds in three steps. First, building on the first part, we address the construction of a random utility representation from observed choice probabilities. From a polyhedral perspective, this construction corresponds to a lifting theorem, which naturally introduces the concepts of the choice branching process and the branching independence property. We then examine the choice branching process of the MNL model and identify its stronger set-independent property. This analysis naturally leads to the introduction of our partial set-dependent random utility model, a flexible intermediate framework that bridges the two extremes.

We also note a direct extension for future work: applying the polyhedral approach to popular limited attention models in economics and operations management, such as the consideration set model [Manzini and Mariotti, 2014], the random attention model [Cattaneo et al., 2020], and the rational consideration model [Akchen and Mitrofanov, 2023].

1.2 ORGANIZATION

This note is structured as follows. Section 2 introduces the random utility model and the necessary polyhedral foundations. Section 3 provides a complete proof of the RUM axiomatization theorem, complementing Fiorini’s argument with a detailed polyhedral analysis. Section 4 then extends this framework to analyze other models, culminating in the introduction of the partial set-dependent RUM. We conclude with future research directions in Section 5.

2 BASIC DEFINITION

2.1 CHOICE MODEL

Let S be a finite set of alternatives. Let $\succ$ denote a strict linear ordering (rank) of S . Let $\mathcal{R}$ be the set of all possible ranks of S . Let $\mu \in \Delta(\mathcal{R})$ denote the distribution of the consumer type such that $\mu(\succ) \geq 0$ for all $\succ \in \mathcal{R}$ and $\sum_{\succ \in \mathcal{R}} \mu(\succ) = 1$. μ measures the probability of the event that the consumer type of rank $\succ$ occurs. Define a **stochastic choice function** as $p : S \times 2^S \rightarrow [0, 1]$ such

that $p(s, T) > 0$ only if $s \in T$ and $\sum_{s \in T} p(s, T) = 1$ for all $T \subseteq S$. $p(s, T)$ measures the probability of choosing s from T . [Block and Marschak, 1959] defines the *random utility model* (or equivalently *rank-based model*) as follows.

Definition 2.1 (Random Utility Model). *Let $N(s, T) := \{\succ \in \mathcal{R} : s \succ t \text{ for all } t \in T \setminus s\}$. A stochastic choice system (S, p) has a rank list representation if there exists a consumer type distribution μ such that*

$$p(s, T) = \sum_{\succ \in N(s, T)} \mu(\succ) \quad \text{for } T \subseteq S \text{ and } s \in S$$

[Falmagne, 1978] gives the axiomatization theorem of random utility model as follows.

Axiom 2.1 (Block and Marschak). *For all $T \subseteq S, s \in T$,*

$$q(s, T) := \sum_{U: T \subseteq U \subseteq S} (-1)^{|U|-|T|} p(s, U) \geq 0. \quad (1)$$

In addition, we call $q(s, T)$ Block-Marschak polynomial.

Theorem 2.1 (Falmagne). *A stochastic choice system has a RU representation if and only if it satisfies Block and Marschak axiom 2.1.*

The proof of necessity part is simple which applies the Möbius inversion, while the proof of sufficiency part is challenging. We gives the necessity part right below and leave the sufficiency part to the section 3. One can also get an intuition from the following proof about what the B-M polynomial stands for.

Proof of Theorem 2.1 (Necessity). Suppose p admits a RU representation. Define a set of ordering,

$$N^*(s, T) := \{\succ \in \mathcal{R} : t' \succ s \succ t, \forall t' \in T^C, t \in T \setminus \{s\}\}.$$

That is, $N^*(s, T)$ contains the ordering such that s is superior to any element inside T (s itself is excluded) and inferior to any element outside T . Thus, the following equation between $N(s, T)$ and $N^*(s, T)$ naturally holds: $N(s, T) = \bigcup_{U: T \subseteq U \subseteq S} N^*(s, U)$. Since p has a RU representation,

$$p(s, T) = \mu(N(s, U)) = \sum_{U: T \subseteq U \subseteq S} \mu(N^*(s, U)).$$

By Möbius inversion,

$$\mu(N^*(s, T)) = \sum_{U: T \subseteq U \subseteq S} (-1)^{|U|-|T|} p(s, U) = q(s, T)$$

Since for RU representation $\mu(N^*(s, T))$ is nonnegative, $q(s, T)$ is nonnegative. $\square$

Remark 2.1. From the proof of the necessity part, we know that in RUM $q(s, T)$ represents the probability that s is superior to any element in T and inferior to any element outside T .

Remark 2.2. q is defined using p as 1. Conversely, we can represent p using q :

$$p(s, T) = \sum_{U: T \subseteq U \subseteq S} q(s, U). \quad (2)$$

[Falmagne, 1978] first gave an algebraic proof of the sufficiency part. In 2004, [Fiorini, 2004] gave a simplification of Falmagne’s theorem, using the combinatoric polytope. In our paper (section 3), we follow his sketch of the proof, while we fully specify the dimensional argument which he omitted. Before we start the sufficiency part, we first introduce the polytope approach.

2.2 POLYTOPES

In this subsection, we see the stochastic choice model from a polytope perspective. We see type distribution μ as a type distribution vector $\boldsymbol{\mu}$. Define $\boldsymbol{\mu} \in \mathbb{R}^{|\mathcal{R}|} = \mathbb{R}^{|S|^!}$, $\boldsymbol{\mu}(\succ) = \mu(\succ)$. Then $\boldsymbol{\mu}$ forms a simplex.

Definition 2.2 (Type Distribution Simplex). *Define the type distribution simplex, denoted by Δ_{type} , as:*

$$\Delta_{type} = \left\{ \boldsymbol{\mu} \in \mathbb{R}^{|\mathcal{R}|} : \boldsymbol{\mu} \geq \mathbf{0}, \sum_{\succ \in \mathcal{R}} \boldsymbol{\mu}(\succ) = 1 \right\}.$$

Let I denote the set of all pairs (s, T) with $s \in T$ and $T \subseteq S$. We see observed choice function p as a choice vector $\mathbf{p}$. Define the choice vector as $\mathbf{p} \in \mathbb{R}^{|I|}$, $\mathbf{p}(s, T) := p(s, T)$. Consider the choice rule of random utility representation as a linear transformation $c : \mathbb{R}^{|\mathcal{R}|} \rightarrow \mathbb{R}^{|I|}$, $c(\boldsymbol{\mu}) = \mathbf{p}$ such that $\mathbf{p}(s, T) = \sum_{\succ \in N(s, T)} \boldsymbol{\mu}(\succ)$. By definition, a p admits random utility representation if and only if its corresponding choice vector $\mathbf{p}$ is in the choice polytope $c(\Delta_{type})$ of RUM. We will show that $c(\Delta_{type})$ is the *rational choice polytope* defined as follows. Denote $\mathbf{p}^\succ$ as the choice vector of a deterministic rank $\succ \in \mathcal{R}$.

$$\mathbf{p}^\succ(s, T) = \begin{cases} 1 & \text{if } s \succ t \text{ for all } s \in T \cup \{0\} \text{ and } T \subseteq S, \\ 0 & \text{otherwise.} \end{cases}$$

We define the $\mathcal{V}$ -representation of the rational choice polytope ($\mathcal{P}_{RC}$)

Definition 2.3 (Rational Choice Polytope). *Define the convex hull of all vectors $\mathbf{p}^\succ$, for $\succ \in \mathcal{R}$ as rational choice polytope. Denote rational choice polytope as $\mathcal{P}_{RC}$.*

From the perspective of polytope theory, a natural question is to characterize the $\mathcal{H}$ -representation of $\mathcal{P}_{RC}$. As implied by 2.1, the characterization is the Block-Marschak linear inequality system. Thus, we convert the axiomatization of RUM to characterizing the $\mathcal{H}$ -representation of $\mathcal{P}_{RC}$.

Next, we introduce the second polytope: choice flow polytope. The choice flow polytope is defined based on the hasse diagram of the lattice $(2^S, \subseteq)$. Formally, a directed graph D is given as follows. Let $D := (V, A)$, where $V = 2^S$ and $A = \{(T \setminus \{s\}, T) : (s, T) \in I\}$. For a $\emptyset$ - S dipath Γ in D , we define a vector $r^\Gamma \in \mathbb{R}^{|A|}$:

$$r(A) = \begin{cases} 1 & \text{if } A \in \Gamma \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.4 (Choice Flow Polytope). Define the convex hull of vectors r^Γ for all $\emptyset$ - S dipath Γ as choice flow polytope. Denote choice flow polytope as $\mathcal{P}_{CF}$

The above definition is the $\mathcal{V}$ -representation of the choice flow polytope. In combinatorial optimization, it is known that the $\mathcal{H}$ -representation of the choice flow polytope can be characterized by the following linear system.

$$r(i, j) \geq 0, \forall (i, j) \in A, \quad (\text{Nonnegative}) \quad (3)$$

$$\sum r(\delta^+(S)) = \sum r(\delta^-(\emptyset)) = 1, \quad (\text{Add-up-to-one}) \quad (4)$$

$$r(\delta^-(v)) = r(\delta^+(v)), \forall v \in N \setminus \{\emptyset, S\}. \quad (\text{Flow Balance}) \quad (5)$$

Finally, we introduce the third polytope: the Block-Marschak polytope ($\mathcal{P}_{BM}$). It is defined by the $\mathcal{H}$ -representation given by the non-negativity constraints on the Block-Marschak polynomials. We will show that $\mathcal{P}_{BM}$ and $\mathcal{P}_{RC}$ are precisely the same polytope, defined by their $\mathcal{H}$ -representation and $\mathcal{V}$ -representation, respectively.

Definition 2.5 (Block Marschak Polytope). Define the Block-Marschak polytope, denoted by $\mathcal{P}_{BM}$, as:

$$\mathcal{P}_{BM} = \left\{ \mathbf{p} \in \mathbb{R}^{|I|} : \begin{array}{ll} \sum_{s \in T} \mathbf{p}(s, T) = 1, & \text{for all } T \subseteq S; \\ \sum_{U: T \subseteq U \subseteq S} (-1)^{|U|-|T|} \mathbf{p}(s, U) \geq 0, & \text{for all } T \subseteq S, s \in T \end{array} \right\}$$

3 POLYHEDRAL PROOF OF THEOREM 2.1

In this section, we will prove the sufficiency part of Falmagne's theorem. We provide a sketch of proof as follows. We first specify the connection between RUM, the rational choice polytope ($\mathcal{P}_{RC}$) and the choice flow polytope ($\mathcal{P}_{CF}$). Step 1 shows the equivalence of the RUM and the rational choice polytope. Thus, we convert the axiomatization of RUM into characterizing $\mathcal{H}$ -representation of $\mathcal{P}_{RC}$. We derive this characterization via the choice flow polytope. Step 2 establishes a one-to-one correspondence between $\mathcal{P}_{RC}$ and $\mathcal{P}_{CF}$ via the linear map f . Step 3 characterizes the $\mathcal{H}$ -representation. To prepare for this step, we prove a flow balance result (Lemma 3.2). Therefore, we complete the proof. The figure 1 is an illustration.

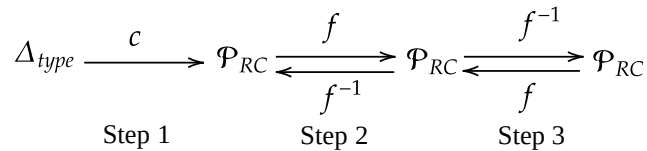


Figure 1: illustration of polyhedral proof of Theorem 2.1

Also, note that our proof differs from the traditional proof structure of axiomatization theorem (or representation theorem) in prior choice model literature. A traditional proof usually follows the following steps:

1. Given that p satisfies the axiom, construct the model parameters and derive the model representation p_{model} ;
2. Verify that the constructed model p_{model} aligns with choice observation p .

In particular, Falmagne’s proof for random utility axiomatization follows such proof structure. In contrast, our polyhedral approach does not necessarily need the model representation construction, as we alternatively study the polytope characterization. However, the connection between the traditional proof and our polyhedral approach will be established in the next section 4.

We first discuss the connections between RUM and the two polytopes defined in the previous section. The following step shows that the RUM choice function can be viewed in the perspective of polytope. For preparation, we introduce the following lemma in convex analysis.

Lemma 3.1. *Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a linear map. Let $\mathcal{P}_1, \mathcal{P}_2 \subset \mathbb{R}^m, \mathbb{R}^n$ be polytopes (i.e. convex hulls of finitely many points). Denote by $\text{Vert}(\mathcal{P})$ the set of vertices of a polytope $\mathcal{P}$. If $f(\text{Vert}(\mathcal{P}_1)) = \text{Vert}(\mathcal{P}_2)$, then $f(\mathcal{P}_1) = \mathcal{P}_2$.*

Step 1 (RUM $\Leftrightarrow$ Rational Choice Polytope). *p can be represented by a random utility model if and only if the corresponding choice vector $\mathbf{p}$ is in $\mathcal{P}_{RC}$.*

Proof of step 1. By definition, p admits RU representation if and only if its corresponding choice vector $\mathbf{p}$ is in $c(\Delta_{type})$. It suffices to show that $c(\Delta_{type}) = \mathcal{P}_{RC}$. We can easily verify that for any vertex $\{\boldsymbol{\mu} \in \Delta_{type} : \boldsymbol{\mu}(\succ) = 1\}$ of Δ_{type} , c maps it to a $\mathbf{p}^\succ$. Thus, $c(\text{Vert}(\Delta_{type})) = \text{Vert}(\mathcal{P}_{RC})$. By lemma 3.1, $c(\Delta_{type}) = \mathcal{P}_{RC}$. $\square$

Next, we show that the $\mathcal{V}$ -representation of $\mathcal{P}_{RC}$ and $\mathcal{P}_{CF}$ is isomorphic. Let f denote the linear transformation given in the B-M polynomial. That is, $f : \mathbb{R}^{|I|} \rightarrow \mathbb{R}^{|A|}$ such that $f(\mathbf{p}) = \mathbf{r}$ with $\mathbf{r}(T \setminus \{s\}, T) = q(s, T)$, where $q(s, T)$ is given by 1. Then, we show that f maps between vertices of two polytopes. In this proof, we visualize that the vertices of $\mathcal{P}_{CF}$ representing a dipath in D are one-to-one corresponding to the vertices of $\mathcal{P}_{RC}$ representing a deterministic rank.

Step 2 ($\mathcal{V}$ -representation $\Leftrightarrow$ Choice Flow Polytope). *$\mathbf{p} \in \mathbb{R}^{|I|}$ is in $\mathcal{P}_{RC}$ if and only if $\mathbf{r} := f(\mathbf{p}) \in \mathbb{R}^{|A|}$ is in $\mathcal{P}_{CF}$.*

Proof of step 2. We first show the “only if” direction. Let $\mathbf{p}^\succ$ be any vertex of $\mathcal{P}_{RC}$. By construction, $\mathbf{p}^\succ$ corresponds to the deterministic ranking $\succ$. Define $\mathbf{r}^\succ := f(\mathbf{p}^\succ)$. For any arc $(T \setminus \{s\}, T) \in A$,

$$\mathbf{r}^\succ(T \setminus \{s\}, T) = \begin{cases} 1, & \text{if } s \text{ is the } \succ\text{-best element in } T \text{ and the } \succ\text{-worst in } T^\complement \cup \{s\}, \\ 0, & \text{otherwise.} \end{cases}$$

That is, $\mathbf{r}^\succ$ sends unit flow exactly along the unique directed path in D that adds alternatives from the worst to the best according to $\succ$. Hence $\mathbf{r}^\succ$ is a feasible $\emptyset$ - S flow and thus $\mathbf{r}^\succ \in \mathcal{P}_{CF}$.

Next, we compare the vertex sets. There are exactly $|S|!$ rankings, so $\mathcal{P}_{RC}$ has $|S|!$ vertices. Likewise, the flow polytope $\mathcal{P}_{CF}$ has one vertex per $\emptyset$ - S directed path in D , and there are exactly $|S|!$ such paths. The mapping f sends each ranking $\succ$ to its corresponding path, hence $f(\text{Vert}(\mathcal{P}_{RC})) = \text{Vert}(\mathcal{P}_{CF})$. Since f is linear, Lemma 3.1 implies $f(\mathcal{P}_{RC}) = \mathcal{P}_{CF}$. This completes the “only if” direction.

For the “if” direction, note that f is a Möbius inversion and therefore invertible. Applying the same argument to f^{-1} yields $f^{-1}(\mathcal{P}_{CF}) = \mathcal{P}_{RC}$. Combining both directions proves the step. $\square$

Remark 3.1. From the proof of 2, we learned that a dipath in the network represents a rank. For example, the red path in figure 2 represents $c \succ b \succ a$. The flow unit distribution of paths represents the probability distribution of deterministic rank types. In addition, we note that an arc represents an alternative in S . For simplicity, let $(s_1, s_2, \dots, s_{|S|})$ where $s_i \in S$ and $s_i \neq s_j, \forall i \neq j$, denote the representative dipath of rank $s_1 \prec s_2 \prec \dots \prec s_{|S|}$.

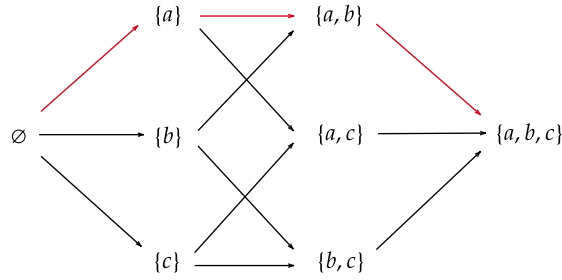


Figure 2: network flow graph for $\{a, b, c\}$

Now, we characterize the $\mathcal{H}$ -representation of rational choice polytope via the choice flow polytope. As shown before, a vector is in choice flow polytope if it satisfies 3 4 5. We provide the following lemma to prepare for the flow balance condition 5. We also emphasized that this flow balance result is omitted in Fiorini’s proof. Thus, our proof will serve as a complement.

Lemma 3.2 (Flow Balance). *Given a stochastic choice function $p(s, T)$ and its corresponding Block-Marschak polynomial $q(s, T)$, we have the following equation.*

$$\sum_{s \in T} q(s, T) = \sum_{s \in S - T} q(s, T \cup \{s\}), \forall T \subseteq S. \quad (6)$$

Sketch of proof of Lemma 3.2. This lemma is from equation 2 and can be proved by induction. In network graph, equation 6 means that the flow units coming in equals the flow units coming out. We illustrate it by an example.

Let $S = \{a, b, c\}$. For each node T , let $\mathcal{T}$ be the set of T and all its succeeding vertices, that is, $\mathcal{T} := \{U : T \subseteq U \subseteq S\}$. Equation 2 can be interpreted as $\mathcal{T} \cdot \mathcal{T}^c$ cut equals to one for all such $\mathcal{T}$. Thus,

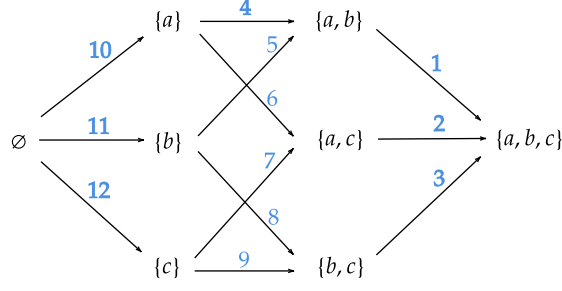


Figure 3: flow balance in a 3 item case

in our example, the following equations holds.

$$\mathbf{1} + \mathbf{2} + \mathbf{3} = 1 \quad T = \{a, b, c\} \quad (7)$$

$$\mathbf{2} + \mathbf{3} + \mathbf{4} + \mathbf{5} = 1 \quad T = \{a, b\} \quad (8)$$

$$\mathbf{1} + \mathbf{3} + \mathbf{6} + \mathbf{7} = 1 \quad T = \{a, c\} \quad (9)$$

$$\mathbf{1} + \mathbf{2} + \mathbf{8} + \mathbf{9} = 1 \quad T = \{b, c\} \quad (10)$$

$$\mathbf{3} + \mathbf{5} + \mathbf{7} + \mathbf{10} = 1 \quad T = \{a\} \quad (11)$$

$$\mathbf{2} + \mathbf{4} + \mathbf{9} + \mathbf{11} = 1 \quad T = \{b\} \quad (12)$$

$$\mathbf{1} + \mathbf{6} + \mathbf{8} + \mathbf{12} = 1 \quad T = \{c\} \quad (13)$$

(7) and (8) give the flow balance equation in vertex $\{a, b\}$: $\mathbf{1} = \mathbf{4} + \mathbf{5}$. Similarly, we derive flow balance equation in vertices $\{a, c\}$ $\{b, c\}$ from (7)(9), (7)(10). Thus, total flow units are passed to the second layer:

$$\mathbf{1} + \mathbf{2} + \mathbf{3} = \mathbf{4} + \mathbf{5} + \mathbf{6} + \mathbf{7} + \mathbf{8} + \mathbf{9} = 1 \quad (14)$$

Inductively, by (11) and flow balance in vertex $\{c\}$, we have $\mathbf{10} = 1 - \mathbf{8} - \mathbf{9} - \mathbf{5} - \mathbf{7}$. Then by applying 14, we have the flow balance equation in vertex $\{a\}$: $\mathbf{10} = \mathbf{4} + \mathbf{6}$. Similarly, based on the flow balance equation on the previous layer and (12)(13), we derive the flow balance equation in vertices $\{b\}, \{c\}$ and the flow balance result in the preceding layer, we can conclude the flow balance in nodes $|T| = 1$. We put the of general case $|S| = N$, $N \in \mathbb{N}$ in the appendix A.1

□

Step 3 (Choice Flow Polytope $\Leftrightarrow \mathcal{H}$ -representation). For a vector $\mathbf{p} \in \mathbb{R}^{|I|}$, $\mathbf{r} := f(\mathbf{p}) \in \mathbb{R}^{|A|}$ is in $\mathcal{P}_{CF}$ if and only if $\mathbf{p}$ is in $\mathcal{P}_{BM}$

Proof of step 3. We first show the “only if” direction. By condition 1, the add-up-to-one condition 4 is automatically satisfied. By condition 1 and proposition 3.2, $f(\mathbf{p})$ satisfies flow-balance condition 5.

By condition 2, nonnegative flow 3 is satisfied. Therefore, $f(\mathbf{p})$ is in $\mathcal{P}_{CF}$. The “if” direction is also satisfied, as f is invertible. Combining this we complete the proof. $\square$

Proof of Theorem 2.1 (Sufficiency). By step 1, (S, p) has RU representation if and only if its choice vector $\mathbf{p} \in \mathbb{R}^{|A|}$ is in $\mathcal{P}_{RC}$. By step 2, it suffices to show that $f(\mathbf{p})$ is in $\mathcal{P}_{CF}$. Since p satisfies B-M conditions, its choice vector $\mathbf{p}$ satisfies conditions in step 3. Then by step 3, $f(\mathbf{p})$ is in $\mathcal{P}_{CF}$. $\square$

Remark 3.2. From the above proof, we know that the flow unit of each arc is given by the B-M polynomial q . The distribution of flow units on the paths represents the probability distribution μ of the deterministic ranking. Recall the discussion of traditional proof at the beginning. For a p that admits a RU representation, we would like to construct μ . This introduces branching-dependent random utility model and we will discuss it in the next section.

4 POLYHEDRAL APPROACH FOR OTHER CHOICE MODELS

In this section, we will apply the polyhedral approach to analyze other choice models, including: branching-independent random utility model [Suleymanov, 2024], multinomial logit model (MNL). Based on the analysis, we then introduce a partial set-dependent random utility model. Similar idea is adapted from [Feng et al., 2023].

Our discussion is structured as follows. In Subsection 4.1, we address two fundamental questions: how to construct the type distribution μ from the choice function p , and whether this construction is unique. This introduces the choice branching process and the branching-independent property of RUMs, while the uniqueness question leads to the issue of model identification. In Subsection 4.2, we examine the choice branching process of the MNL model and identify its set-independent property. This allows us to establish a connection between RUM and MNL through the lens of set-dependence. Building on this connection, subsection 4.3 introduces an intermediate model: the partial set-dependent random utility model, which generalizes both extremes by allowing for an intermediate degree of set-dependence. This model is designed to overcome the practical limitations of both the fully flexible RUM and the highly structured MNL.

4.1 IDENTIFICATION AND BRANCHING-INDEPENDENT RANDOM UTILITY

As mentioned in remark 3.2, for the RUM, we would like to construct the type distribution of the rank μ based on observed data p . We address two main questions: (1) How can μ be constructed? (2) Is this construction unique? From a polyhedral perspective, these questions can be unified under a lifting theorem. As established, the RUM choice function projects the high-dimensional type distribution polytope Δ_{type} to the lower-dimensional random choice polytope $\mathcal{P}_{RC}$: $c(\Delta_{type}) = \mathcal{P}_{RC}$. Given a point $\mathbf{p}$ in $\mathcal{P}_{RC}$, a central question is whether it can be lifted back to a point μ in Δ_{type} that rationalizes it. Furthermore, we aim to characterize the conditions under which such a lifting is unique.

We first discuss the uniqueness of the construction. This is called *Identification* in the economics literature ([Strzalecki, 2025] Section 2.1). A formal definition is given as follows.

Definition 4.1 (Identification). *The RU representation (S, p) can be identified if*

$$\text{for any } \mu' \in \Delta(\mathcal{R}), \text{ if } \mu' \text{ also represents then } \mu' = \mu.$$

It is a well known result that the RUM cannot be uniquely identified in general. However, [Turansick, 2022] characterize the special cases where RU representation of S, p can be identified. As defined in the previous section, let $D := (V, A)$ be its directed network graph and $\mathbf{r} \in \mathbb{R}^{|A|}$ be the flow units assigned to each arc. Define $D' = (N, A')$ as its degenerated graph, where $A' := \{(T \setminus \{s\}, T) \in A : \mathbf{r}(T \setminus \{s\}, T) > 0\}$. That is, D' removes arcs with 0 flow units in D . Let $\delta^+(v) := \{a \in A : v \text{ is the head of } a\}$, $\delta^-(v) := \{a \in A : v \text{ is the tail of } a\}$.

A corridor C in D' is a path with a head v_h and a tail v_t such that

$$\begin{aligned} |\delta^+(v)| &= |\delta^-(v)| = 1, & \forall v \in V(C) \setminus \{v_h, v_t\}; \\ |\delta^+(v_h)| &\geq 2 \text{ or } |\delta^-(v_h)| \geq 2; \\ |\delta^+(v_t)| &\geq 2 \text{ or } |\delta^-(v_t)| \geq 2. \end{aligned}$$

Note that a vertex is always a corridor. Two paths Γ_1, Γ_2 are said to *cross* each other if there exists a corridor $C \subseteq V(\Gamma_1) \cap V(\Gamma_2)$ such that

$$\begin{aligned} A(\Gamma_1) \cap \delta^+(C) &\neq A(\Gamma_2) \cap \delta^+(C), \\ A(\Gamma_1) \cap \delta^-(C) &\neq A(\Gamma_2) \cap \delta^-(C), \end{aligned}$$

In addition, such corridor C is called the *intersection* of Γ_1, Γ_2 . In figure 4, Γ_1, Γ_2 and Γ_3, Γ_4 in the left graph and the middle graph is not intersected. Γ_5, Γ_6 in the right graph is intersected.

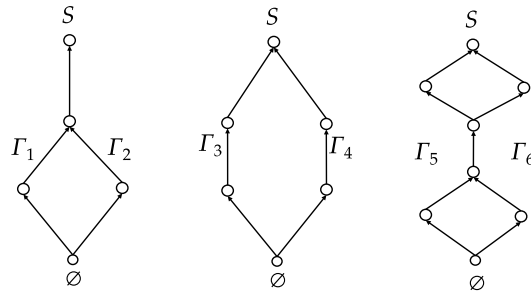


Figure 4: a illustration of intersected paths and non-intersected paths

Proposition 4.1 ([Turansick, 2022]). *The RU representation of (S, p) can be identified if and only if there is no pair of paths in its corresponding degenerated graph D' that cross each other.*

Proof. See Appendix A.2

□

Proposition 4.1 has an intuitive geometric interpretation. As discussed in the previous section, the type distribution μ can be interpreted as the flow distribution in the corresponding network graph D . Whenever there exists an intersection in D , the flow distribution cannot be uniquely determined. Figure 5 is a simple illustration of this case.

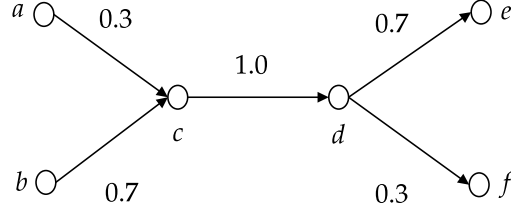


Figure 5: a illustration of Proposition 4.1: in such cross, flow units can be assigned in the following two ways. (1) $b \rightarrow c \rightarrow d \rightarrow e : 0.7$, $a \rightarrow c \rightarrow d \rightarrow f : 0.3$. (2) $b \rightarrow c \rightarrow d \rightarrow e : 0.49$, $b \rightarrow c \rightarrow d \rightarrow f : 0.21$, $a \rightarrow c \rightarrow d \rightarrow f : 0.09$, $a \rightarrow c \rightarrow d \rightarrow e : 0.21$.

Remark 4.1. When there are only three alternatives, RU representation can always be identified, since there are no crosses in the original graph (see figure 2). In addition, for any $\succ \in \mathcal{R}$, say $a \succ b \succ c$, $\mu(\succ)$ is determined by $q(b, \{b, c\})$.

Remark 4.2. [Doignon and Saito, 2023] interpret the identification as the adjacency of vertices in the flow polytope. We know that the identity vectors of the paths in network flow graph are the vertices of flow polytope. [Doignon and Saito, 2023] showed that two vertices of the flow polytope are non-adjacent if and only if their corresponding paths on the graph cross each other. Therefore, proposition 4.1 is equivalent to the following:

Proposition 4.2. The RU representation of (S, p) can be identified if and only if for any two ranks $\succ_1, \succ_2$, such that, if $\mu(\succ_1) > 0$ or $\mu(\succ_2) > 0$, then their corresponding extreme points in the choice flow polytope are adjacent.

The above results answer the question: given the observed choice function p , is the construction of μ uniquely determined. We know that the uniqueness only occurs in special cases characterized by proposition 4.1. Next, we move to the second question: if an observed choice function p admits a random utility representation, how can we find a particular construction μ^* .

Let $s_i(\succ)$ be the i -th best alternative of rank $\succ$. Let $T_i(\succ) := \{s \in S : s_i(\succ) \succ s\} \cup s_i(\succ)$ be a set of alternatives that are not superior to $s_i(\succ)$. The following construction is due to [Falmagne, 1978].

$$\mu^*(\succ) := \prod_{i=1}^{|S|} \frac{q(s_i(\succ), T_i(\succ))}{\sum_{s \in T_i(\succ)} q(s, T_i(\succ))}, \forall \succ \in \mathcal{R}, \quad (15)$$

This construction has a geometric interpretation. For convenience in our analysis, we reverse the direction of all arcs in D and denote it as D_{rev} . Then a S - $\emptyset$ path $(s_1, s_2, \dots, s_{|S|})$ where $s_i \in S$ and $s_i \neq s_j, \forall i \neq j$, represents a rank $s_1 \succ s_2 \succ \dots \succ s_{|S|}$. Let $\Gamma \circ s$ denote the path formed by extending

the path Γ with the arc s , provided that s starts where Γ ends. A path $\Gamma' = \Gamma \circ s$ is said to be a *branched path* of the path Γ . In contrast, Γ is said to be the *merged path* of Γ' . We say that the *branching ratio* from a merged path $\Gamma_i = (s_1, s_2, \dots, s_i)$ to its branched path $\Gamma_{i+1} = \Gamma_i \circ s_{i+1}$ is

$$b(\Gamma_i, s_{i+1}) := \frac{\mathbf{r}(\Gamma_{i+1})}{\mathbf{r}(\Gamma_i)}.$$

From the perspective of choice modeling, the branching ratio measures the conditional probability that alternative s is the next to be ranked, given that the set Γ constitutes the top-ranked alternatives. Formally, for a sequence $s_1, s_2, \dots, s_i$, define $N(s_1 s_2 \dots s_i)$ as the set of ranks in which these alternatives appear as the top i alternatives, and in the given sequential order. That is,

$$N(s_1 s_2 \dots s_i) = \{\succ \in \mathcal{R} : s_1 \succ s_2 \succ \dots \succ s_i \succ t \text{ for all } t \in S \setminus \{s_1, s_2, \dots, s_i\}\}.$$

Then, the branching ratio $b(\Gamma_i, s_{i+1})$ from the path $\Gamma_i = (s_1, \dots, s_i)$ to alternative s_{i+1} is given by the conditional probability:

$$b(\Gamma_i, s_{i+1}) = \frac{\mu(N(s_1 s_2 \dots s_i s_{i+1}))}{\mu(N(s_1 s_2 \dots s_i))}.$$

Next, we describe a choice branching process. Consider a flow of one unit entering at the source vertex S . This flow branches into multiple paths. At each vertices, the incoming flow is subsequently split among the outgoing arcs according to the respective branching ratios. Let $T_i := S \setminus \{s_1, \dots, s_{i-1}\}, \forall i \geq 2$ and $T_1 := S$. Here, in the directed graph of the RUM, the branching ratio from Γ_i to $\Gamma_{i+1} = \Gamma_i \circ s_{i+1}$ is set to

$$b(\Gamma_i, s_{i+1}) = \frac{q(s_{i+1}, T_i)}{\sum_{t \in T_i} q(s, T_i)}, \quad (16)$$

We repeat this branching process until the branched paths reach the terminal vertex $\emptyset$.

More concretely, at the vertex $T_1 = S$, one unit flow branches into paths

$$\mu(N'(s_1)) = \mathbf{r}(\Gamma_1) = q(s_1, T_1), \forall s_1 \in S.$$

In its successor vertices T_2 , the path branches into

$$\mu(N'(s_1 s_2)) = \mathbf{r}(\Gamma_2) = q(s_1, T_1) \cdot \frac{q(s_2, T_2)}{\sum_{s \in T_2} q(s, T_2)}, \forall s_1, s_2 \in S,$$

where $s_1 \neq s_2$. Then, after $k = |S|$ times of branching, we derive complete $\emptyset$ - S paths:

$$\mu(N'(s_1 s_2 \dots s_k)) = \mathbf{r}(\Gamma_k) = \prod_{i=1}^k \frac{q(s_i, T_i)}{\sum_{s \in T_i} q(s, T_i)}, \forall s_1, s_2, \dots, s_k \in S,$$

where $s_i \neq s_j, \forall i \neq j$. By rearranging the notation, these become Falmagne's constructions:

$$\mu(\succ) = \mathbf{r}(\Gamma^\succ) = \prod_{i=1}^{|\mathcal{S}|} \frac{q(s_i(\succ), T_i(\succ))}{\sum_{s \in T_i(\succ)} q(s, T_i(\succ))}, \forall \succ \in \mathcal{R}.$$

We notice that the Falmagne's construction 15 of μ is generated from the path branching process with the branching ratio set to 16.

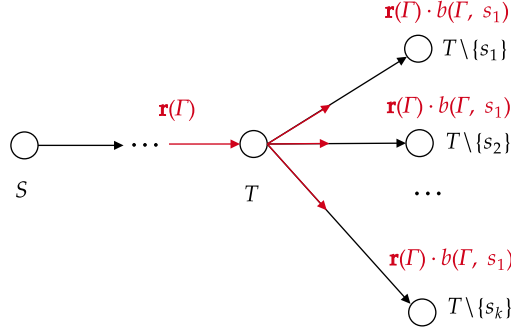


Figure 6: a illustration of paths branching process. A S - T path Γ with flow units of $\mathbf{r}(\Gamma)$ branches in the vertex T . The branching ratio $b(\Gamma, s_i) = q(s_i, T) / \sum_{j=1}^k q(s_j, T)$.

The branching process described above is formulated in terms of a network graph. It also has a choice behavioral interpretation: an individual first identifies their most preferred alternative, s_1 . If s_1 is contained in the offered menu, they choose it. Otherwise, they transit to their second-most preferred alternative according to the branching ratio. This process continues until either an alternative available in the menu is selected, or the least preferred alternative $s_{|\mathcal{S}|}$ is reached.

Note that the branching ratio from a path Γ to an arc s depends solely on the *set* of arcs in Γ , and is independent of their sequence. In the language of random utility theory, this means that given a set $\{s_1, \dots, s_i\}$ of the i most preferred alternatives, the conditional probability of s_{i+1} being the $(i+1)$ -th choice depends only on the set itself, and not on the internal ordering of its elements. This property of branching independence is formally captured in the following proposition.

Proposition 4.3 (Branching Independence (BI)). *When the branching ratio is set to 16, for all vertex $T \subseteq S$, for any two S - T paths Γ, Γ' , then*

$$\frac{b(\Gamma, s_1)}{b(\Gamma, s_2)} = \frac{b(\Gamma', s_1)}{b(\Gamma', s_2)}, \forall s_1, s_2 \in T,$$

Proof. By entering 16,

$$\frac{b(\Gamma, s_1)}{b(\Gamma, s_2)} = \frac{q(s_1, T)}{q(s_2, T)} = \frac{b(\Gamma, s_1)}{b(\Gamma, s_2)}.$$

□

Therefore, [Suleymanov, 2024] introduced a branching-independent random utility model.

Definition 4.2 (Branching-independent Random Utility Model). *A stochastic choice system (S, p) admits a branching-independent RU representation if (S, p) admits a RU representation with a branching-independent μ^* given by 15.*

Theorem 4.1 ([Suleymanov, 2024]). *A stochastic choice system (S, p) has a branching-independent RU representation if and only if it satisfies Block and Marschak axiom 2.1. In addition, the branching-independent RU representation of (S, p) can be identified.*

Proof of Theorem 4.1. From the geometric argument above, we know that every RUM guarantees a corresponding branching-independent RUM through a path branching RUM. Therefore the axiomatization of branching-independent RUM is exactly the same as RUM. Moreover, since μ is uniquely determined by 15, the identification property also holds. $\square$

4.2 MULTINOMIAL LOGIT MODEL AND SET INDEPENDENCE

The Multinomial Logit model (MNL), also known as Luce's model [Luce et al., 1959], is another well-studied stochastic choice model in the economics and operations management literature. It assigns a weight $w(s) > 0$ to each alternative s and defines the choice function $p(s, T)$ for any menu T by a fractional form.

Definition 4.3 (Multinomial Logit Model). *A stochastic choice system (S, p) admits a multinomial logit representation if there exists a function $w : S \rightarrow \mathbb{R}$ such that*

$$p(s, T) = \frac{e^{w(s)}}{\sum_{t \in T} e^{w(t)}}, \forall T \subseteq S, s \in T.$$

It is a well known result that the MNL model is a special case of the RUM. [McFadden, 1974] showed that the MNL representation has an equivalent RUM representation in which the utility function is additively separable into a deterministic utility term and a noise term following the Gumbel distribution. Alternatively, We illustrate this nested relation using the network graph and choice branching process.

Proposition 4.4. *If A stochastic choice system (S, p) has an MNL representation, then it has a RU representation with*

$$\mu(\succ) = \prod_{i=1}^{|S|} p(s_i(\succ), T_i(\succ)). \quad (17)$$

Proof of Proposition 4.4. Suppose (S, p) has a MNL representation v . Then

$$q(s, T) = \sum_{U: T \subseteq U \subseteq S} (-1)^{|U|-|T|} p(s, U) = e^{v(s)} \left(\sum_{U: T \subseteq U \subseteq S} (-1)^{|U|-|T|} \frac{1}{\sum_{t \in U} e^{v(t)}} \right) > 0.$$

Thus, p satisfies B-M axiom and has a RU representation. In addition,

$$\frac{q(s_i(\succ), T_i(\succ))}{\sum_{s \in T_i(\succ)} q(s, T_i(\succ))} = \frac{e^{v(s_i(\succ))}}{\sum_{s \in T_i(\succ)} e^{v(s)}} = p(s_i(\succ), T_i(\succ))$$

By Theorem 4.1, p has a RU representation with branching-independent μ ,

$$\mu(\succ) = \prod_{i=1}^{|S|} \frac{q(s_i(\succ), T_i(\succ))}{\sum_{s \in T_i(\succ)} q(s, T_i(\succ))} = \prod_{i=1}^{|S|} p(s_i(\succ), T_i(\succ)), \forall \succ \in \mathcal{R}.$$

□

Remark 4.3. We note that μ in 17 is constructed by setting the branching ratio to 16. For MNL model, this branching ratio can be further specified. For any vertices $T \subseteq S$ and arcs $s \in T$ in D_{rev} , the branching ratio 16 from a path Γ_i to $\Gamma_i \circ s_{i+1}$ can be specified as

$$b(\Gamma_i, s_{i+1}) = \frac{q(s_{i+1}, T_i)}{\sum_{t \in T_i} q(t, T_i)} = \frac{e^{w(s_{i+1})}}{\sum_{t \in T_i} e^{w(t)}}. \quad (18)$$

Recall the branching independence property 4.3 in RUM. As is shown above, the MNL model demonstrates an enhanced version of the independence property. This is described by the following proposition.

Proposition 4.5 (Set Independence (SI)). *For all $s_1, s_2 \in S$ and arbitrarily two paths Γ, Γ' with end vertices T, T' such that $s_1, s_2 \in T, T'$,*

$$\frac{b(\Gamma, s_1)}{b(\Gamma, s_2)} = \frac{b(\Gamma', s_1)}{b(\Gamma', s_2)} = \frac{e^{v(s_1)}}{e^{v(s_2)}}.$$

The set independence (SI) condition can be interpreted as: the choice branching between s_1, s_2 is independent of its predecessor alternatives in rank. We notice that the set-independent condition is similar to the independence of irrelevant alternatives (IIA) condition.

Axiom 4.1 (Independence of Irrelevant Alternatives (IIA)). *A stochastic choice system (S, p) satisfies IIA axiom if $\forall T, T' \subseteq S, s_1, s_2 \in T \cap T'$,*

$$\frac{p(s_1, T)}{p(s_2, T)} = \frac{p(s_1, T')}{p(s_2, T')}.$$

The IIA condition indicates that the choice between two alternatives s_1, s_2 is independent of the irrelevant alternatives. The IIA condition is known to be a necessary and sufficient condition for the MNL model. We now compare the SI condition with the IIA condition. The SI property is a characterization of the MNL model in RUM parameter spaces μ , while the IIA axiom is a characterization in the choice probability space p .

Next, We compare the SI condition 4.5 in MNL model with the BI condition 4.3 in the RUM model. BI requires P, P' to share the same end vertex, while set independence does not. Thus, SI implies BI.

In a branching process of constructing the complete path (rank) with BI satisfied, to decide what is the probability that s being the next alternative, we must know the set of top alternatives in rank. In some sense, branching independence can be translated as set dependence.

There is much evidence that the MNL model and the RUM are closely related. However, there are limited theories that discuss the models in between RUM and MNL model. Here we offer another perspective that connects the MNL model and the RUM, with the discussion on their independence property. This discussion naturally leads us to think about the models that lie between RUM and MNL. In these models, the branching ratio to the next alternative depends partially on its predecessor alternatives in rank, unlike RUM, where branching ratios are fully dependent on the entire set of predecessors, or MNL, where they are completely independent. We name these models as the *partial set dependence random utility model* and will fully specify them in the following subsection 4.3.

4.3 PARTIAL SET-DEPENDENT RANDOM UTILITY

Before introducing our partial set-dependent random utility model, we first discuss the limitations of the two extreme cases: the general RUM and the specific MNL. These limitations motivate the need for an intermediate model and highlight the practical value of our proposed framework.

For the RUM, the limitations are the following. First, uncovering the RUM parameter requires choice data from all choice menus. Since the branching ratio of RUM is complete set-dependent, we must have the branching ratio parameters at each vertices to derive a full construction of the model. To determine the branching ratio, $p(s, T)$, $s \in T$ is required in all choice menus $T \subseteq S$. For a set of n alternatives, the total size of all $p(s, T)$ is $\sum_{k=1}^n \frac{(k-1)n!}{(n-k)!k!}$. It is untractable in the application to collect such a large scale of data. It is meaningless, as we wish to predict consumers' choice on an unobserved choice menu based on observed data. Second, RUM makes conservative predictions. Since different RU parameters fit the same data, RUM generally predicts a feasible region within the choice probability simplex. Moreover, since RUM explains a rather general choice range, it will correspondingly induce overfitting issues in model estimation.

For the MNL, the problems come from its explanatory power. Its IIA axiom has been challenged by many empirical data. The famous counterexample of MNL model is the Red Bus, Blue Bus paradox [Debreu, 1960]. Moreover, MNL is limited in explaining population choice data, where heterogeneity of consumer types is common. Since IIA condition is a non-convex constraint, the choice range that the MNL model explains is non-convex. That is, a MNL model cannot be decomposed into a mixture (convex combination) of heterogeneous MNL consumer types.

Figure 7 illustrates the explanatory power and prediction of the RUM and MNL models. As evidenced, both models exhibit significant limitations in practical estimation and prediction of consumer choices. They represent two extremes: the MNL model is often overly restrictive, while the fully general RUM can overfit and is computationally challenging. This rationale motivates the introduction of an intermediate framework: the *partial set dependent model*. The following definition is adapted from [Feng et al., 2023].

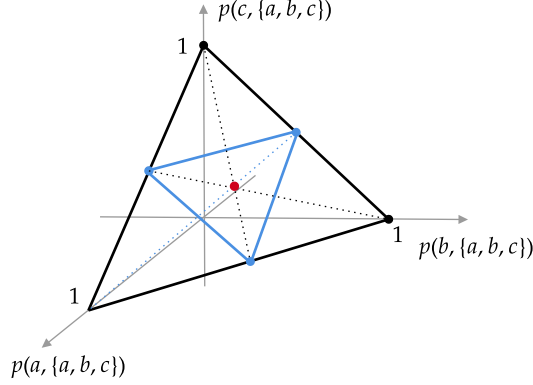


Figure 7: Illustration of the predictions of the RUM and MNL models for $S = \{a, b, c\}$: Suppose we observe binary choice probabilities $p(a, \{a, b\}) = 0.5$, $p(a, \{a, c\}) = 0.5$, $p(b, \{b, c\}) = 0.5$. The black triangle represents the simplex of all possible choice probabilities for the menu $\{a, b, c\}$. the blue region is the set of the RUM prediction. The red dot is the unique MNL prediction $p(a, \{a, b, c\}) = p(b, \{a, b, c\}) = p(c, \{a, b, c\}) = 1/3$.

Let $\Pi(T)$ denote the set of all permutations of the set T . For any permutation $\pi \in \Pi(T)$, it can be viewed as a sequence $(t_1, t_2, \dots, t_n)$ of the elements in T . Denote Γ_π as dipath formed by a sequence of arcs π . For two permutations π and π' , we define their concatenated dipath, denoted by $\Gamma_{\pi, \pi'}$, which represents the dipath formed by traversing Γ_π followed by $\Gamma_{\pi'}$.

Definition 4.4 (Partial set-dependent Random Utility Model). *A subclass of random utility representations is called k -degree set-dependent random utility representation if its corresponding branching ratios satisfy the following:*

$$\frac{b(\Pi_{\pi_1, \pi}, s_i)}{b(\Pi_{\pi_1, \pi}, s_j)} = \frac{b(\Pi_{\pi_2, \pi'}, s_i)}{b(\Pi_{\pi_2, \pi'}, s_j)}, \quad (19)$$

for all $T_0 \subseteq S$, such that $|T_0| = k$, for all $T_1, T_2 \subseteq T_0^c$, for all $\pi, \pi' \in \Pi(T_0)$, $\pi_1 \in \Pi(T_1)$, $\pi_2 \in \Pi(T_2)$.

In addition, if $k = 0$, we call it set independent random utility, if $k = |S| - 1$ we call it set dependent random utility.

In other words, a random utility model exhibits k -degree set dependence if the branching ratio to the next alternative s depends only on the set of its k immediate predecessors in the ranking. As established previously, the MNL model is set-independent (i.e., 0-degree), whereas the RUM is complete set-dependent. The first two parts of the following proposition formalize the degree of set dependence for the MNL model and the RUM, respectively. Furthermore, the higher degree of set dependence implies the lower degree. Consequently, the choice range that can be represented by a RUM with a higher degree of set dependence is nested within the choice range of the lower degree one. This nested structure is formally stated in the third part of the proposition.

Proposition 4.6. *The following propositions hold:*

1. A p admits the MNL representation if and only if it admits a 0-degree set-dependent random

utility representation, or a set-independent random utility.

2. There exists p that admits the $|S| - 1$ -degree set-dependent random utility representation, or a set-dependent random utility representation.
3. For all, $i, j \in \mathbb{N}^+, i < j \leq |S| - 1$. If a p admits a i -degree set-dependent random utility representation, then it also admits a j -degree set-dependent random utility representation.

We now analyze the number of parameters required to specify a k -degree set-dependent RUM. To fully determine a ranking distribution under this model, one can start from the first alternative in the ranking and proceed sequentially, using the branching ratios to determine the next alternative. The k -degree set-dependent RUM is completely determined by the following set of branching ratios $b(\Gamma_\pi, s)$ for all $T \subseteq S$ such that $|T| \leq k$, $s \in T^c$, $\pi \in \Pi(T)$. These parameters, together with the set dependence axiom in Equation 19, suffice to specify the flow units of every S - $\emptyset$ dipaths in D_{rev} , i.e., the distribution of rank types on S . Suppose that $|S| = n$. Then the total number of branching ratios needed is

$$\sum_{i=0}^k (n-i) \cdot \frac{n!}{(n-i)!}.$$

Consequently, the number of parameters grows monotonically with the degree of set dependence. This reveals a trade-off within this model framework: higher model explanatory power come at the cost of increased computational complexity (parameter size).

Note that the degree of set dependence, k , serves as a hyperparameter in our framework. Its value can be tuned on observed choice data to achieve an optimal balance between computational cost and predictive accuracy. This flexibility is particularly valuable in applied settings, where no single model universally dominates. A common strategy in operations management, analogous to a mixture-of-experts (MoE) approach, is to train an ensemble of different models and select the best performer for a given context. We note that [Feng et al., 2023] conducted a numerical experiment using the NielsenIQ Consumer Panel Data. In their study, they adjusted the degree of set dependence to fit the model to the data. For detailed results, we refer the reader to Section 5.3 of [Feng et al., 2023].

5 CONCLUSION REMARKS

We conclude by outlining promising directions for future research, primarily focused on advancing the theory of partial set-dependent random utility models.

First, we aim to investigate the identification problem within this framework. While the fully flexible RUM ($k = |S| - 1$) is generally unidentifiable, and the MNL ($k = 0$) is identified, it remains an open question to characterize the identifiability for intermediate degrees k . A central issue is whether the model becomes identifiable when the parameter space dimension, constrained by k , does not exceed that of the observable choice space.

Second, we seek to establish an axiomatic characterization—that is, a complete description of the

choice range—for partial set-dependent RUMs. While the characterizations for the two extreme cases (RUM and MNL) are well-known, the axiomatization for intermediate models remains an important unsolved problem.

REFERENCES

- [Akchen and Mitrofanov, 2023] Akchen, Y.-C. and Mitrofanov, D. (2023). Consider or choose? the role and power of consideration sets. *arXiv preprint arXiv:2302.04354*.
- [Block and Marschak, 1959] Block, H. D. and Marschak, J. (1959). Random orderings and stochastic theories of response.
- [Cattaneo et al., 2020] Cattaneo, M. D., Ma, X., Masatlioglu, Y., and Suleymanov, E. (2020). A random attention model. *Journal of Political Economy*, 128(7):2796–2836.
- [Chambers et al., 2024] Chambers, C. P., Masatlioglu, Y., and Turansick, C. (2024). Correlated choice. *Theoretical Economics*, 19(3):1087–1117.
- [Debreu, 1960] Debreu, G. (1960). Individual choice behavior: A theoretical analysis.
- [Doignon and Saito, 2023] Doignon, J.-P. and Saito, K. (2023). Adjacencies on random ordering polytopes and flow polytopes. *Journal of Mathematical Psychology*, 114:102768.
- [Falmagne, 1978] Falmagne, J.-C. (1978). A representation theorem for finite random scale systems. *Journal of Mathematical Psychology*, 18(1):52–72.
- [Feng et al., 2023] Feng, Q., Shanthikumar, J. G., and Xue, M. (2023). Rational choice models: The temporal tree representation. *Available at SSRN 4653687*.
- [Fiorini, 2004] Fiorini, S. (2004). A short proof of a theorem of falmagne. *Journal of mathematical psychology*, 48(1):80–82.
- [Luce et al., 1959] Luce, R. D. et al. (1959). *Individual choice behavior*, volume 4. Wiley New York.
- [Manzini and Mariotti, 2014] Manzini, P. and Mariotti, M. (2014). Stochastic choice and consideration sets. *Econometrica*, 82(3):1153–1176.
- [McFadden, 1974] McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In Zarembka, P., editor, *Frontiers in Econometrics*, pages 105–142. Academic press, New York.
- [Strzalecki, 2025] Strzalecki, T. (2025). Stochastic choice theory. *Cambridge Books*.
- [Suleymanov, 2024] Suleymanov, E. (2024). Branching-independent random utility model. *Journal of Economic Theory*, 220:105880.
- [Turansick, 2022] Turansick, C. (2022). Identification in the random utility model. *Journal of Economic Theory*, 203:105489.

A PROOF

A.1 PROOF OF LEMMA 3.2

Proof. We follow a similar induction step and extend the proof to the general case. For $n = 1$ and layer $|T| = |S| - n$, denote $\{t\} = S - T$. By equation 2,

$$\begin{aligned}
& \sum_{s \in T} \sum_{T \subseteq U \subseteq S} q(s, U) = 1 \\
& \Rightarrow \sum_{s \in S - \{t\}} q(s, S - \{t\}) + \sum_{s \in S - \{t\}} q(s, S) = 1 \\
& \Rightarrow \sum_{s \in S - \{t\}} q(s, S - \{t\}) + 1 - q(t, S) = 1 \\
& \Rightarrow \sum_{s \in S - \{t\}} q(s, S - \{t\}) = q(t, S).
\end{aligned}$$

By induction hypothesis, we assume flow balance result holds for $1, 2, \dots, k$ th layer. Since the total flow units in layer 0 equal to one, we can conclude the total flow units equal to one for layer $1, 2, \dots, k$, i.e.

$$\sum_{U: |U|=|S|-n} \sum_{s \in U} q(s, U) = 1, \quad \forall n = 1, 2, \dots, k.$$

For $n = k + 1$, $|T| = |S| - n$, by equation 2

$$\begin{aligned}
& \sum_{s \in T} \sum_{T \subseteq U \subseteq S} q(s, U) = 1 \\
& \Rightarrow \sum_{s \in T} q(s, T) + \sum_{U \in U_1} \sum_{s \in T} q(s, U) + \dots + \sum_{U \in U_{|S|-|T|}} \sum_{s \in T} q(s, U) = \sum_{U: |U|=|S|-k} \sum_{s \in U} q(s, U),
\end{aligned}$$

where $U_1 = \{U \in \mathcal{T} : |U| = |T| + 1\}, \dots, U_{|S|-|T|} = \{U \in \mathcal{T} : |U| = |S|\}$.

$$\begin{aligned}
& \Rightarrow \sum_{s \in T} q(s, T) + \sum_{U \in U_1} \sum_{s \in T} q(s, U) + \dots + \underbrace{\sum_{U \in U_{|S|-|T|-1}} \sum_{s \in T} q(s, U) + \sum_{U: U=S-t, t \in T} \sum_{s \in U} q(s, U)}_{\text{aggregate this two terms}} \\
& = \sum_{U: |U|=|S|-k} \sum_{s \in U} q(s, U) \\
& \Rightarrow \sum_{s \in T} q(s, T) + \sum_{U \in U_1} \sum_{s \in T} q(s, U) + \dots + \sum_{U: |U|=|S|-2, U \cap T = \emptyset} \sum_{s \in S-U} q(s, U \cup \{s\}) = \sum_{U: |U|=|S|-k} \sum_{s \in U} q(s, U).
\end{aligned}$$

By applying induction hypothesis, we have

$$\sum_{s \in T} q(s, T) + \sum_{U \in U_1} \sum_{s \in T} q(s, U) + \cdots + \sum_{U: |U|=|S|-2, U \cap T = \emptyset} \sum_{s \in U} q(s, U) = \sum_{U: |U|=|S|-k} \sum_{s \in U} q(s, U).$$

Through this step, we are using flow balance hypothesis to transfer the arc flow units of lower layers to upper layer. Surprisingly, we find that we can aggregate those arcs in upper layer, which covers exactly all of the flow coming out of node $\{U : U \cap T = \emptyset\}$. Thus, we can apply the flow balance hypothesis repeatedly and iterate this step. Eventually, all arc flow units are transferred to layer $\{U : |U| = |T| + 1 = |S| - k\}$. Then we have the following,

$$\sum_{s \in T} q(s, T) + \sum_{U: |U|=|T|, U \cap T = \emptyset} \sum_{s \in S-U} q(s, U \cup \{s\}) = \sum_{U: |U|=|T|+1} \sum_{s \in U} q(s, U).$$

By moving the second term on the left hand side to the right hand side and reorganizing the equation, we derive the flow balance result for $k+1$ th layer:

$$\sum_{s \in T} q(s, T) = \sum_{s \in S-T} q(s, T \cup \{s\})$$

□

A.2 PROOF OF PROPOSITION 4.1

Proof. (Sufficiency): Let $\Gamma_{\succ}$ be the path that represents the rank $\succ$. By the proof of Theorem 2.1, we know that the flow unit of each path represents the probability distribution of ranking, i.e., $\mu(\succ) = \mathbf{r}(\Gamma_{\succ}), \forall \succ \in \mathcal{R}$.

Suppose that there are no pairs of paths in D' that cross each other. Then for any path $\Gamma_{\succ}$, there exists an arc $a \in V(\Gamma)$ such that $a \notin V(\Gamma'), \forall \Gamma'$ is a path in $D', \Gamma' \neq P$. Let $a_{\succ}$ be such arc for path $\Gamma_{\succ}$. Then $\mu(\succ)$ is uniquely determined by $\mathbf{r}(a)$.

(Necessity): Suppose that (S, p) can be identified. It suffices to show that there are no pairs of paths in D' that cross each other. Prove by contradiction.

Suppose that there exists such a pair Γ_1, Γ_2 . Let $\succ_1, \succ_2$ be their corresponding ranks, respectively. Let C with a tail v_t and a head v_h be their intersection. Then both Γ_1 and Γ_2 are segmented into three paths $\Gamma_1^{\emptyset-v_t}, C, \Gamma_1^{v_h-S}$ and $\Gamma_2^{\emptyset-v_t}, C, \Gamma_2^{v_h-S}$. Now consider two new paths generated by the combinations of above segments: $\Gamma_3 = \Gamma_1^{\emptyset-v_t} \cup C \cup \Gamma_2^{v_h-S}$ and $\Gamma_4 = \Gamma_2^{\emptyset-v_t} \cup C \cup \Gamma_1^{v_h-S}$.

Let μ be a probability distribution of ranks that represents (S, p) . Now we can construct a new μ' as follows:

$$\begin{aligned} \mu'(\succ_1) &= \frac{\mu(\succ_1)}{\mu(\succ_1) + \mu(\succ_2)} \cdot \mu(\succ_1), \\ \mu'(\succ_2) &= \frac{\mu(\succ_2)}{\mu(\succ_1) + \mu(\succ_2)} \cdot \mu(\succ_2), \\ \mu'(\succ_3) &= \mu(\succ_3) + \frac{\mu(\succ_2)}{\mu(\succ_1) + \mu(\succ_2)} \cdot \mu(\succ_1), \\ \mu'(\succ_4) &= \mu(\succ_4) + \frac{\mu(\succ_1)}{\mu(\succ_1) + \mu(\succ_2)} \cdot \mu(\succ_2). \end{aligned}$$

For the path segment $\Gamma_1^{\emptyset-v_t}$, we can check the flow unit that passed through remains the same with new μ' , i.e.,

$$\mathbf{r}(\Gamma_1^{\emptyset-v_t}) = \mu(\succ_1) + \mu(\succ_2) + C = \mu'(\succ_1) + \mu'(\succ_2) + C,$$

where C represents the total flow units of other paths that pass through $\Gamma_1^{\emptyset-v_t}$. It can be easily checked that the same argument holds for other path segments. Thus, μ, μ' form exactly the same directed graph with flow unit assignment. Therefore, both μ and μ' represent (S, p) . However, we assume (S, p) can be identified. A contradiction occurs. $\square$